Grade: REVIEW_REQUIRED

Review Notes:
x Qa: Review manually.
x Qb: Review manually.
x Qc: Review manually.
x Qd: Review manually.
x Qe: Review manually.

Phebe GRAY Quiz 1 2/24/2026

① a) $P(X \geq 60)$ At least 60. 60, 80, 100

$P(X \geq 60) = P(60) + P(80) + P(100)$

$0.15 + 0.25 + 0.05 = 0.45$

The probability that a randomly selected custoner withdraws at least \$60 from the ATM is 45%.

② b) $E[X] = \sum x \cdot P(X = x)$

Average = (value) · (probability) + (value) · (probability) + ...

Amount X	Probability	Multiply
10	.10	$10 \times .10 = 1$
20	.15	$20 \times .15 = 3$
40	.30	$40 \times .30 = 12$
60	.15	$60 \times .15 = 9$
80	.25	$80 \times .25 = 20$
100	0.05	$100 \times .05 = 5$

Sum = $1 + 3 + 12 + 9 + 20 + 5 = 50$ $E[X] = 50$

$SD = \sqrt{E(X^2) - (E[X])^2}$

$E[X^2] = 10 + 60 + 480 + 540 + 1600 + 500 = 3190$

mean = 50

$3190 - 50^2 = 3190 - 2500 = \underline{690}$ variance

$\sqrt{690} = 26.27$

③ c) The Expected Value is an average of thousands of withdraws. it's a theorectial /statistical amount. a long-run average. it does not mean someone may withdraw \$50. it's not a guranteed individual outcome. If we repeat this situation many times, the average would approch that number it's a weighted average. but not a prediction of a guranteed outcome.

④ d)

Mean = 50, SD ≈ 26.27

One SD from the mean:

$50 - 26.27 \leq X \leq 50 + 26.27$

$23.73 \leq X \leq 76.27$

$P(23.73 \leq X \leq 76.27) = P(40) + P(60)$

$= 0.30 + 0.15$

$= 0.45$

The probability that the amount withdrawn from ATM will fall with one standard deviation of the mean is 45%